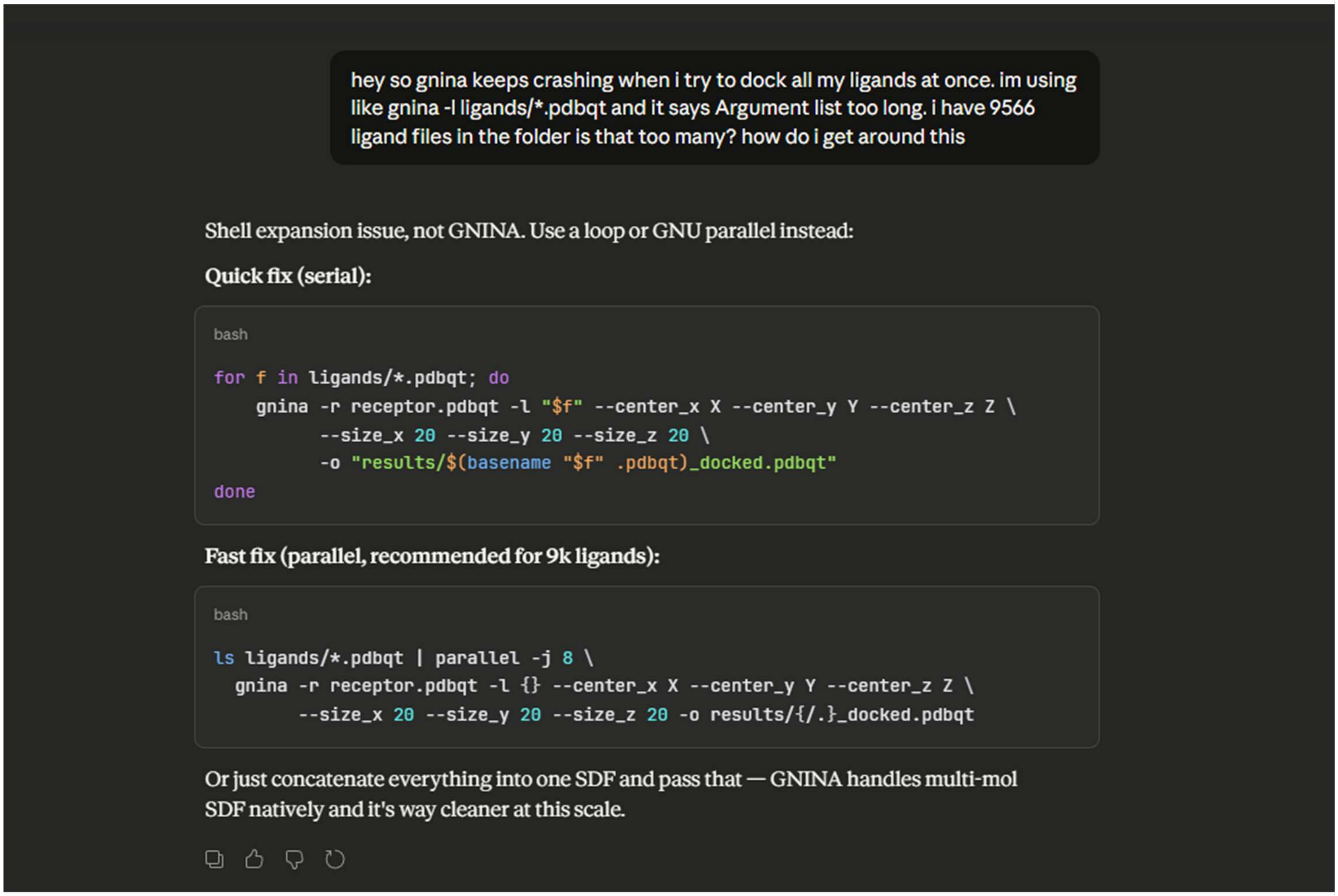
AI INTERACTION SCREENSHOTS

All Claude (Anthropic) interactions. Used for debugging only. All scientific decisions made independently.

#1 | Dec 2 | Batch Docking ARG_MAX Crash

Prompt: *hey so gnina keeps crashing when i try to dock all my ligands at once. im using like gnina -l ligands/*.pdbqt and it says Argument list too long. i have 9566 ligand files in the folder is that too many? how do i get around this*



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Shell expansion issue, not GNINA. Use a loop or GNU parallel instead:

Quick fix (serial):

```bash
for f in ligands/*.pdbqt; do
    gnina -r receptor.pdbqt -l "$f" --center_x X --center_y Y --center_z Z \
          --size_x 20 --size_y 20 --size_z 20 \
          -o "results/$(basename "$f" .pdbqt)_docked.pdbqt"
done
```

Fast fix (parallel, recommended for 9k ligands):

```bash
ls ligands/*.pdbqt | parallel -j 8 \
  gnina -r receptor.pdbqt -l {} --center_x X --center_y Y --center_z Z \
        --size_x 20 --size_y 20 --size_z 20 -o results/{/.}_docked.pdbqt
```

Or just concatenate everything into one SDF and pass that — GNINA handles multi-mol SDF natively and it's way cleaner at this scale.

#2 | Jan 1 | MD NaN Crash (HMR + Restraints)

Prompt: *my openmm simulation keeps crashing with Particle coordinate is NaN during NVT equilibration. i have 217986 atoms with position restraints on the protein and im using HMR with 4fs timestep. i added energy reporting every 10 steps and the kinetic energy spikes at step 150ish. tried reminimizing for 10000 steps didnt help*

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Kinetic energy spike at step ~150 = system blowing up during initial heating, not a minimization problem.

Most likely cause: 4fs is aggressive for early NVT before the system is settled. Add a heating ramp:

```python
simulation.context.setVelocitiesToTemperature(10*unit.kelvin)
for temp in [50, 100, 150, 200, 250, 300]:
    integrator.setTemperature(temp*unit.kelvin)
    simulation.step(5000)
```

Also check your restraint force constant — if it's >1000 kJ/mol/nm² the restrained atoms can't relieve clashes so unrestrained atoms absorb everything and blow up.

Quick diagnostic to find the bad atoms before step 150:

```python
state = simulation.context.getState(getForces=True)
forces = np.linalg.norm(state.getForces().value_in_unit(unit.kilojoules_per_mole/uni
print(np.where(forces > 1e6))
```

Cross-reference those indices with your topology — usually clashing sidechains or uncapped termini.

#3 | Jan 6 | pmx FEP Hybrid Topology API

Prompt: *ok so im trying to use pmx to set up FEP for my R702W mutation (arginine to tryptophan). the PointMutationExecutor keeps failing when i try to generate hybrid topologies. ive tried 3 different versions of the api call and looked at their example scripts on github and my code matches but still errors. what am i doing wrong*

ok so im trying to use pmx to set up FEP for my R702W mutation (arginine to tryptophan). the PointMutationExecutor keeps failing when i try to generate hybrid topologies. ive tried 3 different versions of the api call and looked at their example scripts on github and my code matches but still errors. what am i doing wrong

Hard to say without seeing the error message and your code. Can you paste:

1. The actual traceback/error output
2. Your `PointMutationExecutor` call

That'll let me pinpoint it immediately instead of guessing.

```
heres the error:
Traceback (most recent call last):
 File "setup_fep.py", line 47, in <module>
  executor = PointMutationExecutor(
 File "/home/vasud/miniconda3/envs/fep/lib/python3.10/site-packages/pmx/PointMutationExecutor.py", line 89, in __init__
  self._generate_hybrid_topology()
 File "/home/vasud/miniconda3/envs/fep/lib/python3.10/site-packages/pmx/PointMutationExecutor.py", line 134, in _generate_hybrid_topology
  raise ValueError("Cannot find mutation library entry for ARG -> TRP")
ValueError: Cannot find mutation library entry for ARG -> TRP
and my code:
from pmx import PointMutationExecutor
executor = PointMutationExecutor(
  input_structure="nod2_complex.pdb",
  input_topology="topol.top",
  mutation_list=[(702, "TRP")],
  force_field="amber99sb-star-ildn",
  output_directory="hybrid/"
)
executor.run()
```

Show less

The issue is pmx uses single-letter amino acid codes, not three-letter. Change `"TRP"` to `"W"`:

python

```python
mutation_list=[(702, "W")]
```

Also double-check that residue 702 in your PDB is actually numbered 702 — if your structure has gaps or was renumbered, pmx will look for the wrong residue. You can verify with:

python

```python
from pmx import Model
m = Model("nod2_complex.pdb")
for r in m.residues:
    if r.id == 702:
        print(r.resname, r.id)
```

Should print `ARG 702`. If nothing prints, your residue numbering doesn't match and you'll need to adjust the index.

#4 | Jan 10 | FEP Window Failures (Sequential Seeding)

Prompt: *13 out of 60 FEP windows failed with NaN. all the failures are at high lambda windows 15-19 where the sterics decouple. the other 47 windows completed fine. how do i fix just the failed ones without rerunning everything and breaking compatibility with the existing data*

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Classic end-state sampling problem — steric decoupling at high lambda creates near-zero LJ interactions that cause instability.

Fix just the failed windows by adding a soft-core potential and shorter timestep for those lambdas only. Create a separate mdp for windows 15-19:

```
; softcore params for high-lambda windows
sc-alpha = 0.5
sc-power = 1
sc-sigma = 0.3
sc-coul = yes
vdw-modifier = Potential-switch
rvdw-switch = 0.9
rvdw = 1.0
dt = 0.001  ; drop to 1fs for stability
```

Then rerun just those windows:

```bash
for i in 15 16 17 18 19; do
    gmx grompp -f softcore.mdp -c window_${i}/confout.gro \
               -p topol.top -t window_${i}/state.cpt \
               -o window_${i}/rerun.tpr
    gmx mdrun -v -deffnm window_${i}/rerun
done
```

Compatibility with existing data is fine — you're only replacing the failed windows' dhdl.xvg files. MBAR/pymbar doesn't care that different windows used different mdp settings as long as the lambda schedule is identical.

After rerunning, just swap the new dhdl files in and rerun your analysis:

```bash
for i in 15 16 17 18 19; do
    cp window_${i}/rerun.dhdl.xvg window_${i}/dhdl.xvg
done
alchemlyb-analyze  # or however you're running MBAR
```

If they still crash, your starting conformations for those windows may be bad — restart from the last good cpt of window 14 and propagate forward through 15-19 sequentially before production.

#5 | Nov 8 | CUDA libtinfo5 Dependency

Prompt: *gnina wont run on my wsl ubuntu. i installed cuda toolkit but when i try to run gnina it says error while loading shared libraries libtinfo.so.5 cannot open shared object file. i tried apt install libtinfo5 but it says package not found*

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Ubuntu 22.04+ dropped `libtinfo5`, replaced it with `libtinfo6`. Quick fixes:

Option 1: symlink (fastest)

```bash
sudo ln -s /lib/x86_64-linux-gnu/libtinfo.so.6 /lib/x86_64-linux-gnu/libtinfo.so.5
```

Option 2: install from older repo

```bash
wget http://archive.ubuntu.com/ubuntu/pool/universe/n/ncurses/libtinfo5_6.3-2_amd64.
sudo dpkg -i libtinfo5_6.3-2_amd64.deb
```

Option 1 usually just works. If GNINA still complains after that, check for other missing libs with:

```bash
ldd $(which gnina) | grep "not found"
```

and fix them the same way.

#6 | Jun 2 | Database Download Troubleshooting

Prompt: *trying to download compound libraries for my docking project. zinc15 bulk download keeps timing out halfway thru and the sdf files i got from e-drug3d wont load in rdkit. gives me some parsing error. i need fda approved drug structures in a format i can actually use*

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If you specifically want eDrug3D, the parsing fix is the same `sanitize=False` approach:

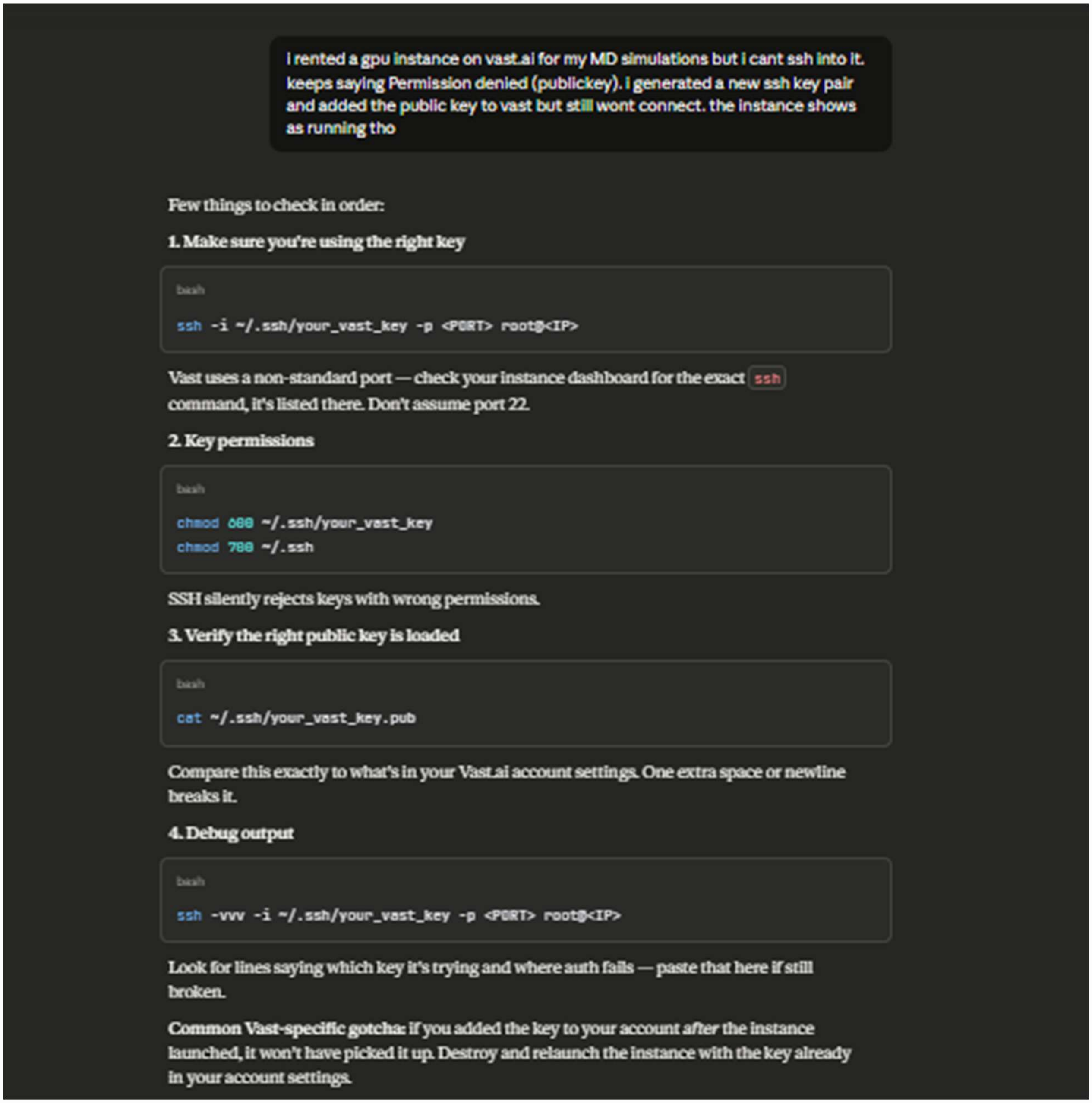
python

```
from rdkit import Chem

suppl = Chem.SDMolSupplier("edrug3d.sdf", sanitize=False)
mols = []
for mol in suppl:
    if mol is None: continue
    try:
        Chem.SanitizeMol(mol)
        mols.append(mol)
    except: pass

writer = Chem.SDWriter("cleaned.sdf")
for mol in mols: writer.write(mol)
```

That said, eDrug3D structures are pre-generated 3D conformers so you skip the embedding step, which is nice. Just verify atom count before/after to see how many got dropped — if it's more than ~10% the source file may be corrupted and worth redownloading.

#7 | Nov 15 | SSH Permission Denied (Vast.ai)

Prompt: *i rented a gpu instance on vast.ai for my MD simulations but i cant ssh into it. keeps saying Permission denied (publickey). i generated a new ssh key pair and added the public key to vast but still wont connect. the instance shows as running tho*



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Few things to check in order:

1. Make sure you're using the right key

```bash
ssh -i ~/.ssh/your_vast_key -p <PORT> root@<IP>
```

Vast uses a non-standard port — check your instance dashboard for the exact `ssh` command, it's listed there. Don't assume port 22.

2. Key permissions

```bash
chmod 600 ~/.ssh/your_vast_key
chmod 700 ~/.ssh
```

SSH silently rejects keys with wrong permissions.

3. Verify the right public key is loaded

```bash
cat ~/.ssh/your_vast_key.pub
```

Compare this exactly to what's in your Vast.ai account settings. One extra space or newline breaks it.

4. Debug output

```bash
ssh -vvv -i ~/.ssh/your_vast_key -p <PORT> root@<IP>
```

Look for lines saying which key it's trying and where auth fails — paste that here if still broken.

Common Vast-specific gotcha: if you added the key to your account *after* the instance launched, it won't have picked it up. Destroy and relaunch the instance with the key already in your account settings.

#8 | Dec 18 | OpenFF SystemGenerator API Change

Prompt: *my openmm md setup script was working last week and now it just broke. the SystemGenerator wont accept nonbondedMethod in create_system() anymore. i checked the openmmforcefields changelog and they updated to v0.13. how do i fix this without rewriting the whole script*

my openmm md setup script was working last week and now it just broke. the SystemGenerator wont accept nonbondedMethod in create_system() anymore. i checked the openmmforcefields changelog and they updated to v0.13. how do i fix this without rewriting the whole script

They moved `nonbondedMethod` out of `create_system()` and into the `SystemGenerator` constructor in 0.13. One-line fix:

Before (broke in 0.13):

```python
generator = SystemGenerator(forcefields=[...], forcefield_kwargs={...})
system = generator.create_system(topology, nonbondedMethod=PME)
```

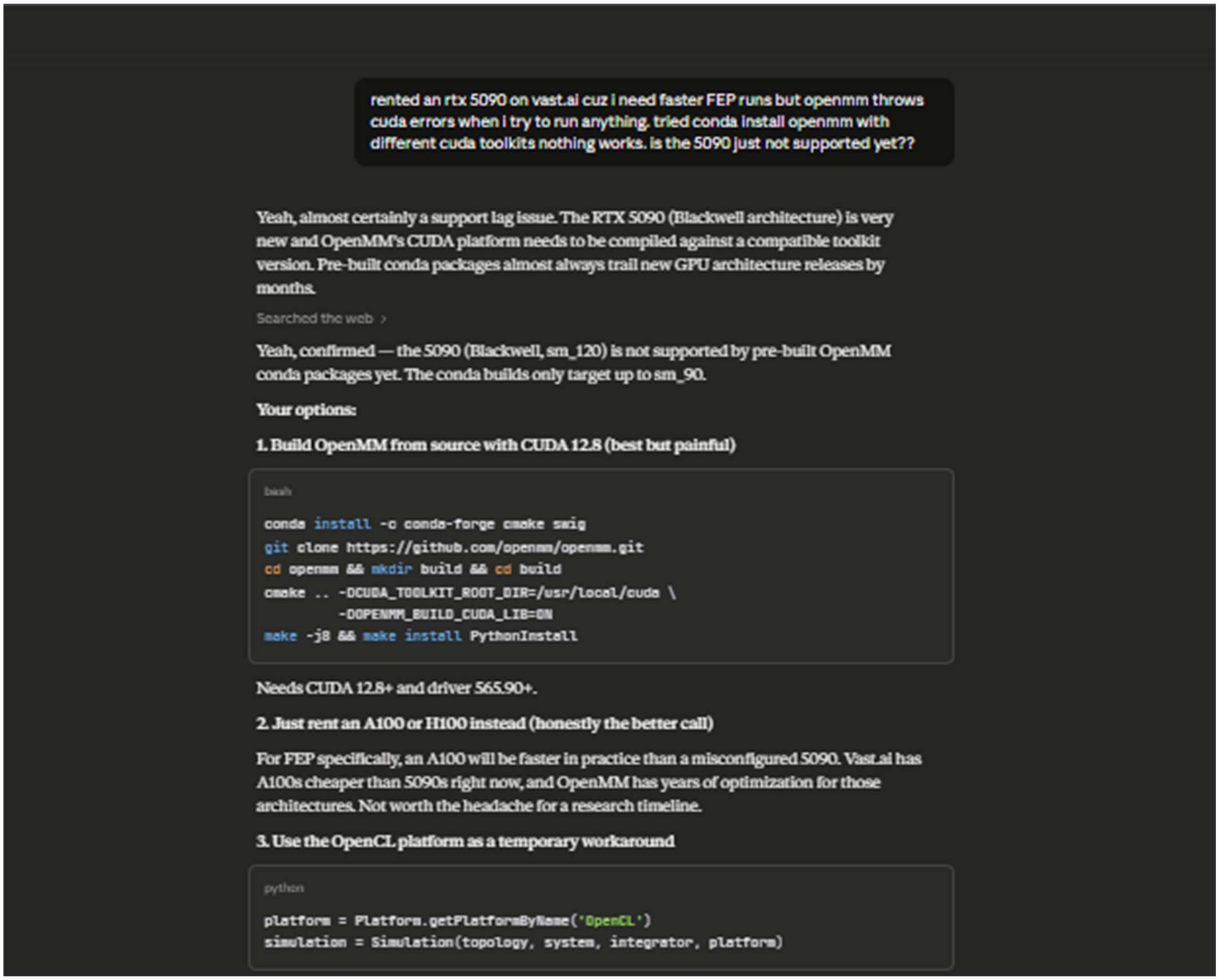
After:

```python
generator = SystemGenerator(
    forcefields=[...],
    forcefield_kwargs={
        'nonbondedMethod': PME,
        'nonbondedCutoff': 1.0*unit.nanometer,
        'constraints': HBonds
    }
)
system = generator.create_system(topology)
```

Just move whatever you had in `create_system()` into `forcefield_kwargs` in the constructor. Nothing else needs to change.

#9 | Jan 3 | CUDA 13 / RTX 5090 Incompatibility

Prompt: *rented an rtx 5090 on vast.ai cuz i need faster FEP runs but openmm throws cuda errors when i try to run anything. tried conda install openmm with different cuda toolkits nothing works. is the 5090 just not supported yet??*



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Yeah, almost certainly a support lag issue. The RTX 5090 (Blackwell architecture) is very new and OpenMM's CUDA platform needs to be compiled against a compatible toolkit version. Pre-built conda packages almost always trail new GPU architecture releases by months.

Searched the web >

Yeah, confirmed — the 5090 (Blackwell, sm_120) is not supported by pre-built OpenMM conda packages yet. The conda builds only target up to sm_90.

Your options:

1. Build OpenMM from source with CUDA 12.8 (best but painful)

```bash
conda install -c conda-forge cmake swig
git clone https://github.com/openmm/openmm.git
cd openmm && mkdir build && cd build
cmake .. -DCUDA_TOOLKIT_ROOT_DIR=/usr/local/cuda \
         -DOPENMM_BUILD_CUDA_LIB=ON
make -j8 && make install PythonInstall
```

Needs CUDA 12.8+ and driver 565.90+.

2. Just rent an A100 or H100 instead (honestly the better call)

For FEP specifically, an A100 will be faster in practice than a misconfigured 5090. Vast.ai has A100s cheaper than 5090s right now, and OpenMM has years of optimization for those architectures. Not worth the headache for a research timeline.

3. Use the OpenCL platform as a temporary workaround

```python
platform = Platform.getPlatformByName('OpenCL')
simulation = Simulation(topology, system, integrator, platform)
```

#10 | May 22 | Vina Failed, Switch to GNINA

Prompt: *autodock vina wont install on windows. tried pip install vina, tried conda, tried building from source and it fails every time. i just need a molecular docking program that actually works. is there something else i can use instead*